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CGU ODISHA

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Department of Chemistry

BIOLOGY AND ENVIRONMENTAL SCIENCE (CY202)

UNIT-III : Structure and Function of Ecosystem

Dr. Gulmi Chakraborty
Assistant Professor

Outline

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Unit III Structure and function of Ecosystem

Unit IV Air and Noise Pollution: Sources, cause and treatment

Unit V Water Pollution Abatement and Solid Waste Management

Environmental Sustainability

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- Human wellbeing is closely linked to the health of the environment.
- People need clean air to breathe, fresh water to drink, and places to live that are free of toxic substances and hazards.
- Environmental sustainability is the responsibility to conserve natural resources and protect global ecosystems to support health and wellbeing, now and in the future.
- As we begin to experience the long-term consequences of exponential industrial growth and energy use, we must act to reverse these effects and prevent further damage, ensuring we have healthy places to live for generations to come.
- Because so many decisions that impact the environment are not felt immediately, a key element of environmental sustainability is its forward-looking nature.

Economic Growth and Environmental Sustainability

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- Companies clearly have a responsibility to society to implement environmentally sustainable practices.
- We now know unrestricted consumption takes a significant toll on human welfare.
- As GDP climbs, so too does our energy use, leading to more polluted environments and depleted natural resources.
- But, that doesn't mean businesses can't be successful and sustainable. Some growth works in step with sustainability.
- Renewable energy companies represent a source of new jobs.
- Using less energy and plastic in production represents an opportunity to grow profit margins.

https://www.boldbusiness.com/energy/solar-energy-vs-fossil-fuels-dominating-industry/

https://www.newindianexpress.com/business/2022/jul/06/single-use-plastic-ban-to-have-minimal-impact-on-big-fmcg-companies-2473447.html

- Environmental scientists and engineers deals with the environmental issues such as global warming, stratospheric ozone depletion, and local and regional air and water pollution.
- Through this course, we train undergraduate and graduate students to think about environmental processes in an integrated fashion, preparing them to manage the environmental challenges we face.
- **'Namami Gange Programme'**, is an Integrated Conservation Mission, approved as 'Flagship Programme' by the Union Government in June 2014 with budget outlay of Rs.20,000 Crore to accomplish the twin objectives of effective abatement of pollution, conservation and rejuvenation of National River Ganga.

https://biology-for-all.weebly.com/lesson-4-effects-of-mans-activities-on-the-environment.html

https://www.theguardian.com/environment/2016/jan/07/human-impact-has-pushed-earth-into-the-anthropocene-scientists-say

- Accordingly, environmental engineering approaches to solutions are fundamental to maintaining healthy environmental systems; and to reducing public health hazards, such as contamination-control, -treatment, and -prevention (e.g., water, air, soil), for hazard exposure reduction.
- In doing so, environmental engineering studies contribute to solving public health challenges by producing valuable analytical insights that engage public health prevention management; and workable engineering solutions to reduce disease and prevent hazard exposure for human wellbeing.

Environment and Ecosystem

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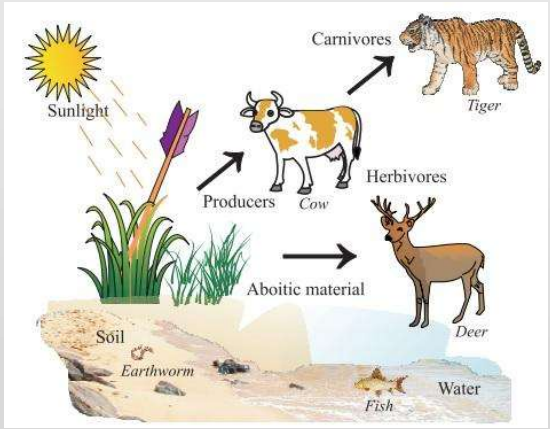
- The environment is the surrounding in which living organisms live. The environment is everything around us, including living organisms and nonliving beings such as soil, air, water, etc.
- The environment helps nourish life on Earth.
- The ecosystem is an organized system of organisms in which living organisms interact with the environment.
- Environment refers to the area where organisms live but the ecosystem is the organized community of living beings where biotic and abiotic elements are in continuous interaction with each other.

https://www.vedantu.com/biology/difference-between-environment-and-ecosystem

Ecosystem

- Ecosystem is the major ecological unit.
- It has both structure and function.
- The structure is related to species diversity.
- In other words, the ecosystem is the basic functional unit of organism and their environment interacting with each other.
- The function of ecosystem is related to the energy flow, decomposition, nutrient cycling and major biomes.

https://agriculturistmusa.com/ecosystem-ecology-how-ecosystems-work/

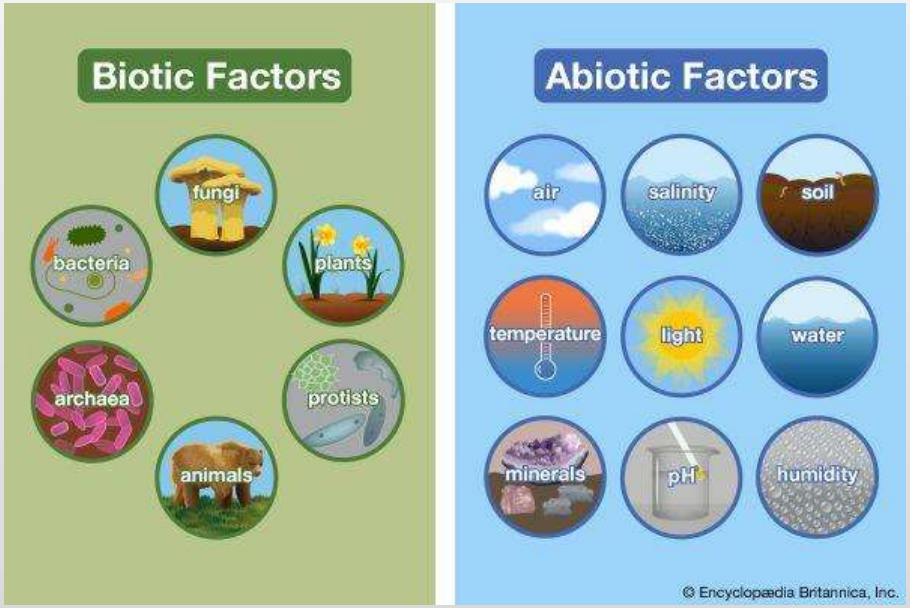


https://www.studypage.in/general-science/ecosystem-and-its-components

Structure of Ecosystem

Generally ecosystems consist of two basic components.

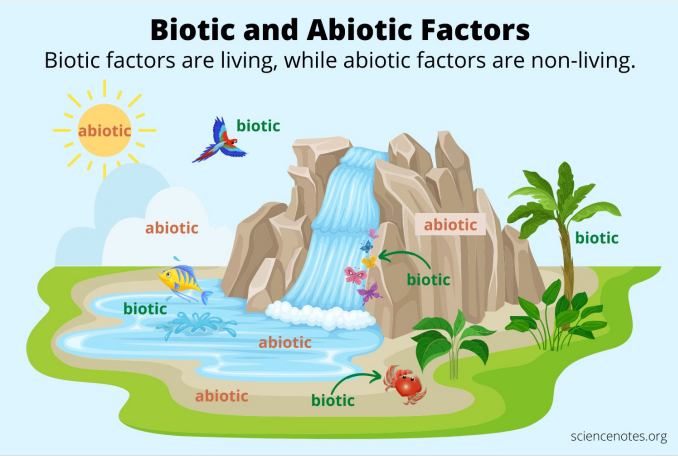
1. Abiotic component.
2. Biotic component.



Abiotic Components

➢It includes basic in-organic (soil, water, oxygen, calcium carbonates, phosphates etc.) and organic compounds.

➢It also includes physical factors such as moisture, wind currents and solar radiation. Radiant energy of sun is the only significant energy source for any ecosystem.



Biotic Components

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Biotic components

Include producers, consumers and decomposers.

Producer: These are the autotrophic, chlorophyll-bearing organisms such as plants, which produce their own food.

Consumers: A consumer which gets nutrition by eating plants is called **Primary consumers** (herbivore) (eg) Rabbit, deer and cow.

The Secondary Consumer: (carnivores) is an animal that eats the flesh of herbivores (eg) cats and dogs.

Tertiary Consumers: are the type of carnivores, which prey upon other carnivores. (eg) Lion, tiger and vulture.

Decomposers

Decomposers attack the dead remains of producers and consumers and degrade the complex organic substances into simpler compounds to derive their nutrients. The decomposers play very important role in maintaining the dynamic nature of ecosystem.

Levels of Organization in The Biotic Component of The Environment

Individual:

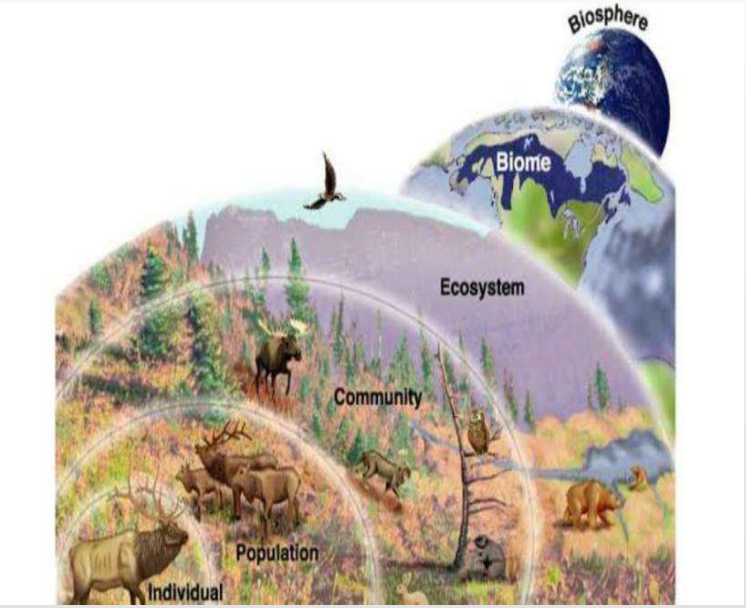
➢All the individuals potentially able to interbreed with one another and produce fertile offspring. An example of a species is the salmon Salmo salar.

Population:

➢Consists of a group of individuals of the same species living in a particular area at the same time.

➢Each population is genetically distinct to some degree from other separate populations of the same species.

➢They have a size and a birth, death and hence population growth rate.



https://civilaspirant.in/levels-of-organisation-in-ecology/

Levels of Organization in The Biotic Component of The Environment

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Community:

- Populations of different species live together, many interacting with each other, forming a community.
- In a pond- a natural community of plants, animals and microbes forming a distinctive living system.
- These interactions lead to the formation of food webs, a hierarchy of who eats who.
- Communities occur in habitats, which refers to the kind of physical environment or place determined by the topography, vegetation structure, geology and surrounding medium (air or water), e.g. forest habitat, seashore habitat etc.
- Some species live in just one habitat, e.g. fish in a lake.
- Others may use several different habitats, e.g. bird species like the crow use fields, seashore, hedgerows, etc.

Levels of Organization in The Biotic Component of The Environment

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Ecosystem:

- This encompasses both the living (biotic) and non-living (abiotic) components of an area- a combination of the community and physical and chemical components of the local environment.
- The major feature is strong interaction between the biotic and abiotic components.
- Major processes are nutrient cycling and energy flow.
- Communities occur in habitats, which refers to the kind of physical environment or place determined by the topography, vegetation structure

Levels of Organization in The Biotic Component of The Environment

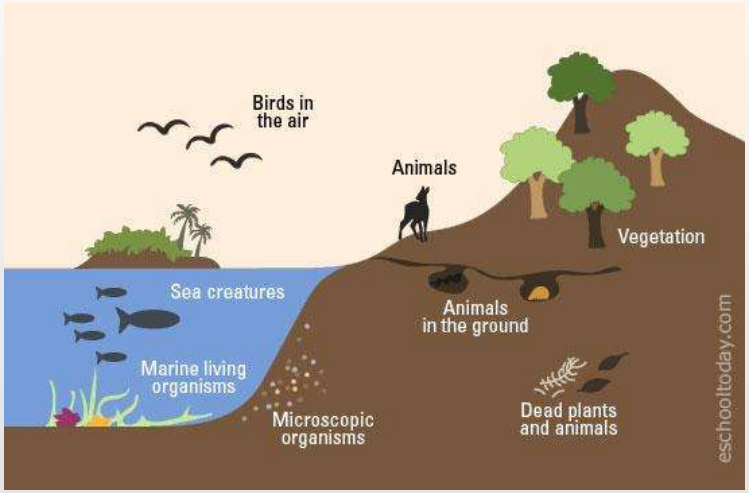
Biome:

- Biome can be defined as a group of organisms that share common characteristics, and these characteristics can be correlated to the environment in which they exist.
- A biome consists of both plants and animals.
- The characteristics of the organisms living inside a biome have developed in response to the environmental conditions.
- Biomes may span more than one continent.
- Biome is a broader term than habitat and can comprise a variety of habitats.
- Examples: tropical rainforest (high temperature and rainfall), coniferous forest or Taiga (low winter temperatures), grassland (warm temperatures, fairly low rainfall), desert scrub(high temperature, very low rainfall).

Levels of Organization in The Biotic Component of The Environment

Biosphere:

- The highest organizational level is the biosphere- that part of the earth and atmosphere in which life exists.
- It includes the surface layer of land, the oceans and sediments at the bottom of water bodies, and part of the atmosphere occupied by life.
- The ultimate aim of the ecological study is to understand how the biosphere functions, in order to be able to predict what affects certain human activities will have in the future and how to ameliorate the problems already caused.



https://eschooltoday.com/learn/the-biosphere/

Ecosystem Processes (Energy Flow and nutrient cycling)

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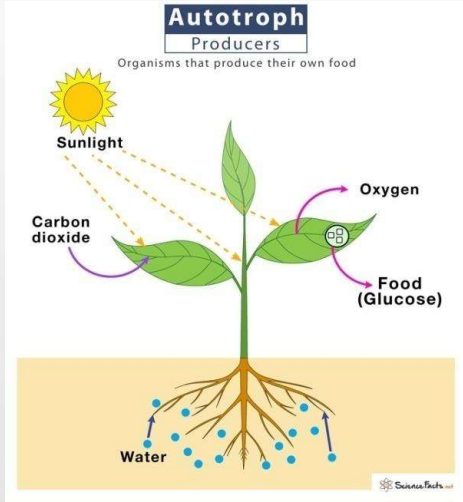
Energy Flow

- In ecosystems, a proportion of the light energy is converted to chemical energy.
- In the environment there are essentially two sources of energy: autotrophic and heterotrophic.
- Autotrophic production of energy rich organic matter is carried out within the ecosystem by green plants in the presence of light via photosynthesis.
- In heterotrophic energy source, the chemical energy is imported as organic matter which originated from primary production in some other ecosystem.
- An example is in heavily shaded forest streams, which are dependent on organic matter, such as dead leaves, entering the stream from the surrounding catchment and carried down from upstream.



https://www.sciencefacts.net/autotrophs-primary-producer.html

https://www.linkedin.com/pulse/how-groundwater-influences-europes-surface-waters-vince-kaandorp

Ecosystem Processes (Energy Flow and nutrient cycling)

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Photosynthesis

➢All green plants synthesize energy-rich organic molecules such as glucose containing chemical energy through the following reactions:

$12\ H_2O + 6\ CO_2 + 709\ kCal \longrightarrow C_6H_{12}O_6 + 6O_2 + 6\ H_2O$

➢Of the total amount of solar radiation, only 1-5% is used in photosynthesis.

➢The glucose produced by plants simply be stored as an energy rich substance in the form of starch or to be combined with other sugar molecules to form carbohydrates like cellulose.

➢With other inorganic substances, like nitrogen, phosphorous, magnesium and iron, the plant produces complex substances like fats, proteins and nucleic acids.

➢The production of organic matter by plants is called **primary production.**

Primary Production

- **Gross primary production** is the total amount of chemical energy (or biomass) stored by plants per unit area per unit time.
- Some of the gross primary production is used in the process of respiration.
- The remaining production, net primary production, can then be used in plant growth and reproduction.
- Thus, **Net primary production** = gross primary production-respiration
- The net primary production is 80-90 per cent of gross primary production levels.

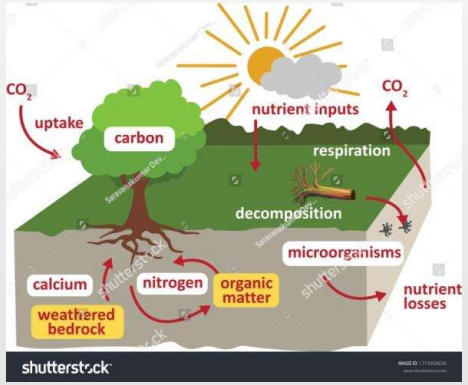
Nutrient Recycling

➢Nutrients are the substances that are required by organisms for growth and development.

➢A nutrient is defined as any substance that provides nourishment to the body and is essential for normal growth and development, reproduction, and survival of the organism.

➢These nutrients are continuously used by the organisms and are added back to the environment in some modified form. The modified form is again converted back to its original form by some other organisms.

➢This cyclic flow of nutrients from the environment to organisms and back into the environment is called nutrient recycling.



https://www.shutterstock.com/search/nutrient-cycling

https://alevelbiology.co.uk/notes/nutrient-recycling/

Energy Flow

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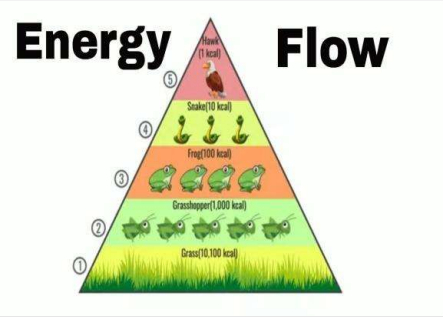
- Ecological energetics is the study of the movement of energy through an ecosystem.
- The sun provides all the energy that living beings require.
- Green plants absorb around 1% of total radiant energy for distribution across the ecosystem.
- Despite its small size, this quantity is adequate to maintain all life on Earth.
- The flow of power that occurs along a food chain is called energy flow.
- Fuel enters the food chain at the level of the producers in the form of solar energy.
- The plants convert solar energy into chemical energy through the process of photosynthesis.
- This chemical energy is passed from one trophic level to the next trophic level along a food chain or from one trophic level to another.



https://www.youtube.com/watch?v=FknsfZnmjQ0

Energy Flow

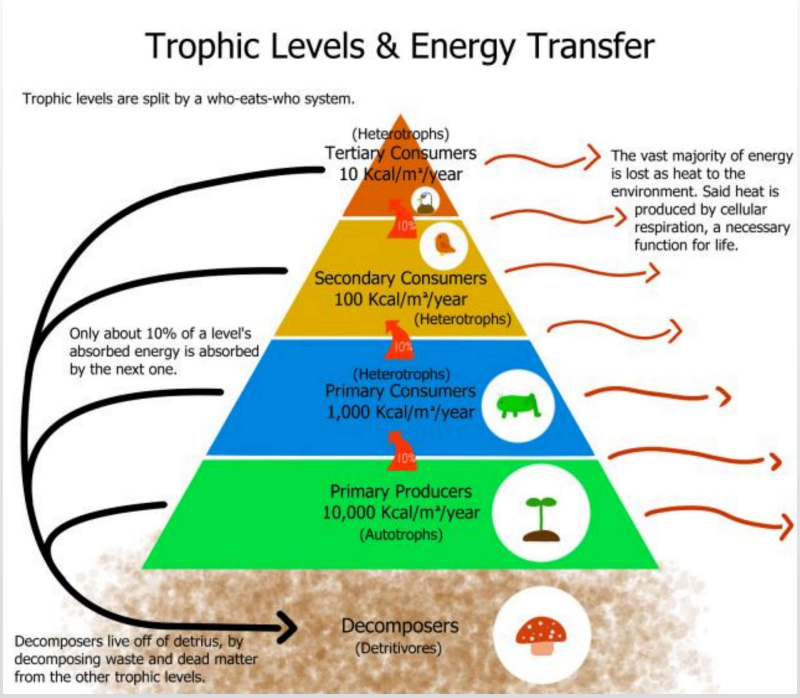
There is a direct relationship between energy flow and thermodynamics Energy flow is based on two different laws of thermodynamics:

(i) First Law of Thermodynamics

The first law of thermodynamics states that energy can neither be created nor destroyed. Here also, the source of energy, i.e., solar energy, can neither be created nor destroyed. It can only be transferred from one system to another as from one form to another form.

(ii) Second Law of Thermodynamics

In the second law, it is stated that during the transformation, a portion of the energy is dissipated into the surroundings as heat energy.



https://en.wikipedia.org/wiki/Energy_flow_%28ecology%29

Food Chain

➢The food chain is the straight chain of organisms that starts with the producer and then the energy is transferred from the producer to the carnivores and ends with the decomposer in the food web.

➢The food chain explains the feeding relationship of organisms.

➢Those who produce their own food by changing the organic energy into chemical energy are known as autotrophs and in the process known as photosynthesis that organisms have chlorophyll for making food. Thus, they are called Producers or Autotrophs.

➢Some organisms which are dependent on the producers as a source of food are termed, **consumers**. Consumers are also known as **herbivores**.

The food chain has four main components those are:

Food Chain

Sun: Plants use solar energy for the process of photosynthesis.

Producer: These are the organisms that produce food themselves through the process of photosynthesis. Producers are also known as **Autotrophs**. Solar energy is converted into chemical energy which transfers from one trophic level to the other as a source of energy. Producer Example-Algae, green plants, cyanobacteria, etc.

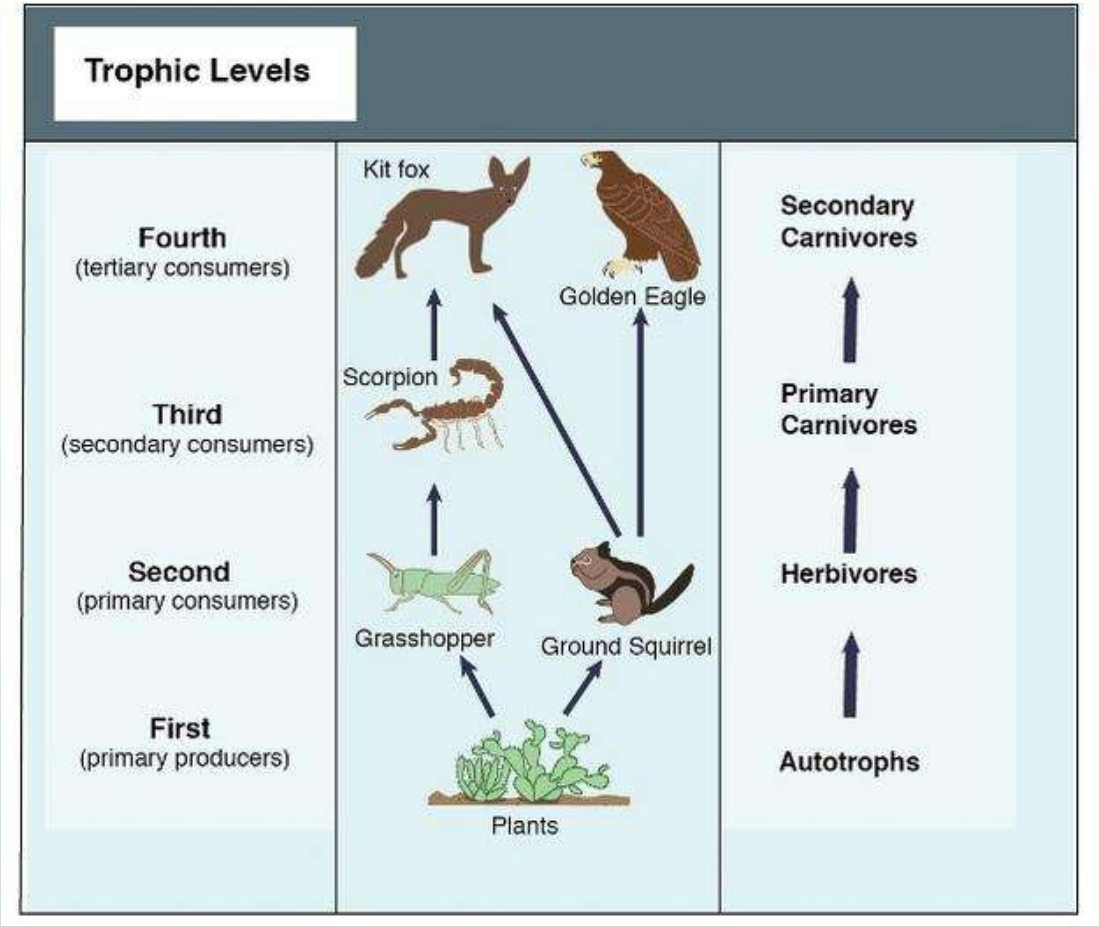
Consumer: These are the organisms that consume the food produced by the producers, also known as **heterotrophs**. Heterotrophs are those organisms that cannot make their own food and depends on other for food. The consumer can be **herbivores**, i.e., that organism that feeds upon green plants, or carnivores i.e., organisms that feed upon another organism. Consumers are the largest part of any food chain.

Decomposers: Organisms that break down complex organic substances into simple inorganic substances, which then go into the soil and they are an important part of maintaining the nutrients cycle in the environment.

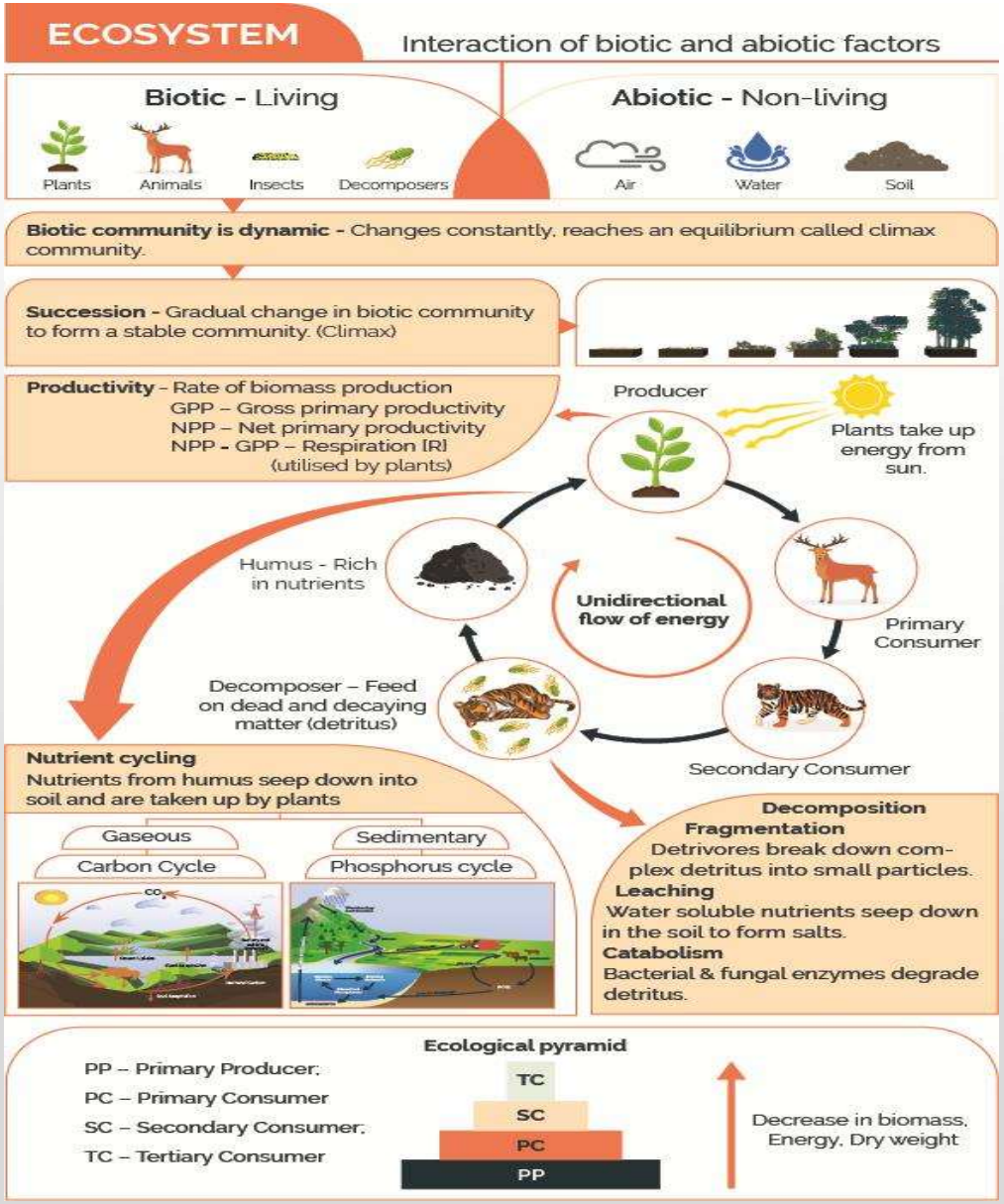
Food Chain

https://www.geeksforgeeks.org/food-chains-and-food-webs/

Food Chain and Food Web



https://www.nature.com/scitable/knowledge/library/food-web-concept-and-applications-84077181/


ECOSYSTEM
Interaction of biotic and abiotic factors
Biotic - Living
Plants
Animals
Insects
Decomposers
Abiotic - Non-living
Air
Water
Soil
Biotic community is dynamic - Changes constantly, reaches an equilibrium called climax community.
Succession - Gradual change in biotic community to form a stable community. (Climax)
Productivity - Rate of biomass production
GPP – Gross primary productivity
NPP – Net primary productivity
NPP - GPP – Respiration [R]
(utilised by plants)
Producer
Plants take up energy from sun.
Humus - Rich in nutrients
Unidirectional flow of energy
Primary Consumer
Decomposer – Feed on dead and decaying matter (detritus)
Secondary Consumer
Nutrient cycling
Nutrients from humus seep down into soil and are taken up by plants
Gaseous
Carbon Cycle
Sedimentary
Phosphorus cycle
Decomposition
Fragmentation
Detrivores break down complex detritus into small particles.
Leaching
Water soluble nutrients seep down in the soil to form salts.
Catabolism
Bacterial & fungal enzymes degrade detritus.
Ecological pyramid
PP – Primary Producer;
PC – Primary Consumer
SC – Secondary Consumer;
TC – Tertiary Consumer
TC
SC
PC
PP
Decrease in biomass, Energy, Dry weight

Biogeochemical Cycles

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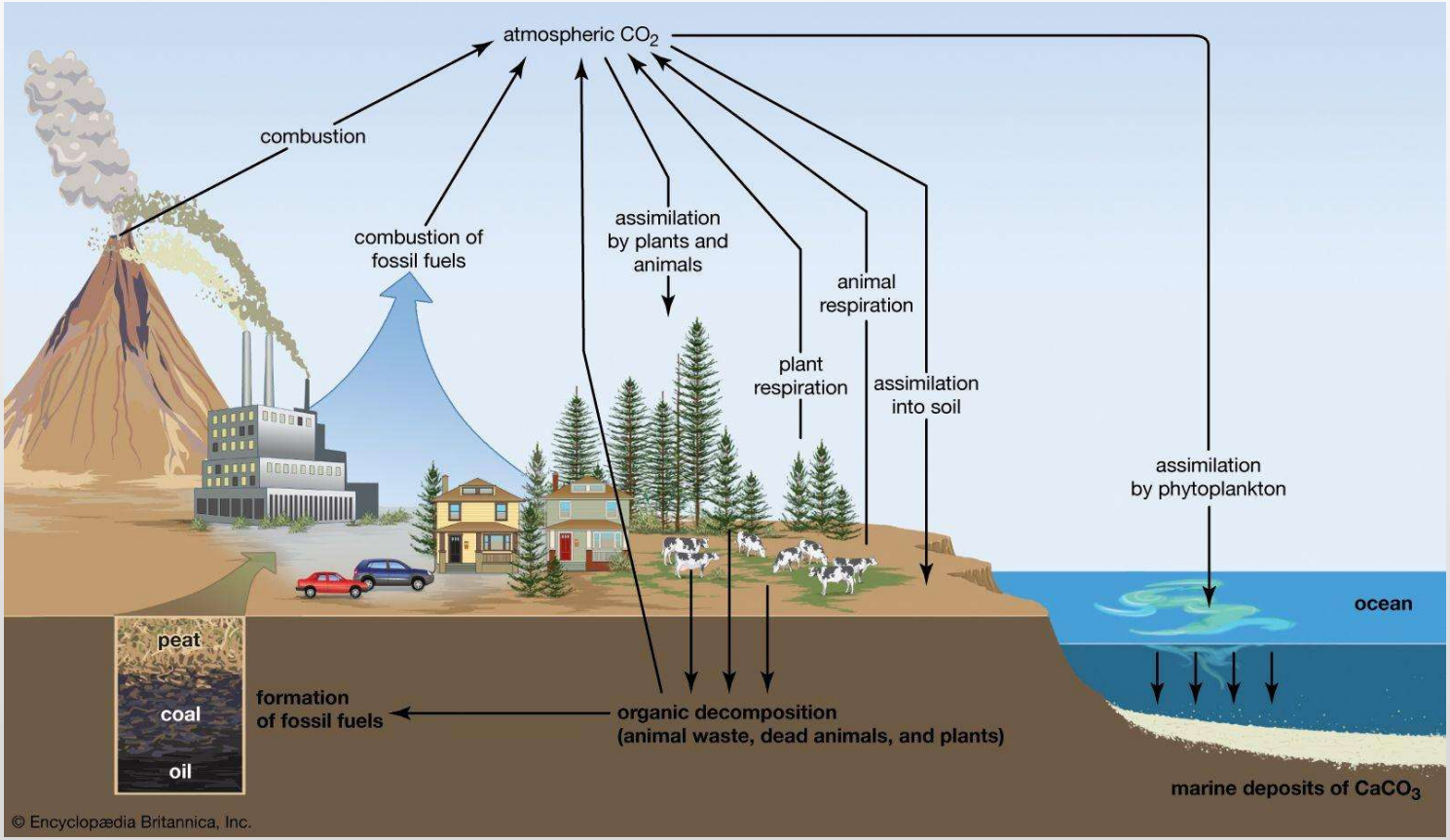
- **Biogeochemical cycle**: essential elements of living matter are circulated.
- The term *biogeochemical* is a contraction that refers to the consideration of the biological, geological, and chemical aspects of each cycle.
- Elements within biogeochemical cycles flow in various forms from the nonliving (abiotic) components of the biosphere to the living (biotic) components and back.

Carbon Cycle



https://www.britannica.com/science/biogeochemical-cycle

Carbon Cycle

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- The **carbon cycle** is that part of the biogeochemical cycle by which carbon is exchanged among the biosphere, pedosphere, geosphere, hydrosphere, and atmosphere of Earth.
- Carbon is the main component of biological compounds as well as a major component of many minerals.
- The carbon cycle comprises a sequence of events that are key to making Earth capable of sustaining life.
- It describes the movement of carbon as it is recycled and reused throughout the biosphere, as well as long-term processes of carbon sequestration (storage) to and release from carbon sinks.

Carbon Cycle

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- Human activities have disturbed the fast carbon cycle for many centuries by modifying land use, and moreover with the recent industrial-scale mining of fossil carbon (coal, petroleum, and gas extraction, and cement manufacture) from the geosphere.
- Carbon dioxide in the atmosphere had increased nearly 52% over pre-industrial levels by 2020, forcing greater atmospheric and Earth surface heating by the Sun.
- The increased carbon dioxide has also caused a reduction in the ocean's pH value and is fundamentally altering marine chemistry.
- The majority of fossil carbon has been extracted over just the past half century, and rates continue to rise rapidly, contributing to human-caused climate change.

Carbon Cycle

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- Carbon in the Earth's atmosphere exists in two main forms: carbon dioxide and methane.
- Both of these gases absorb and retain heat in the atmosphere and are partially responsible for the greenhouse effect.
- Methane produces a larger greenhouse effect per volume as compared to carbon dioxide, but it exists in much lower concentrations and is more short-lived than carbon dioxide.
- Thus, carbon dioxide contributes more to the global greenhouse effect than methane.
- Carbon dioxide is removed from the atmosphere primarily through photosynthesis and enters the terrestrial and oceanic biospheres.

Carbon Cycle

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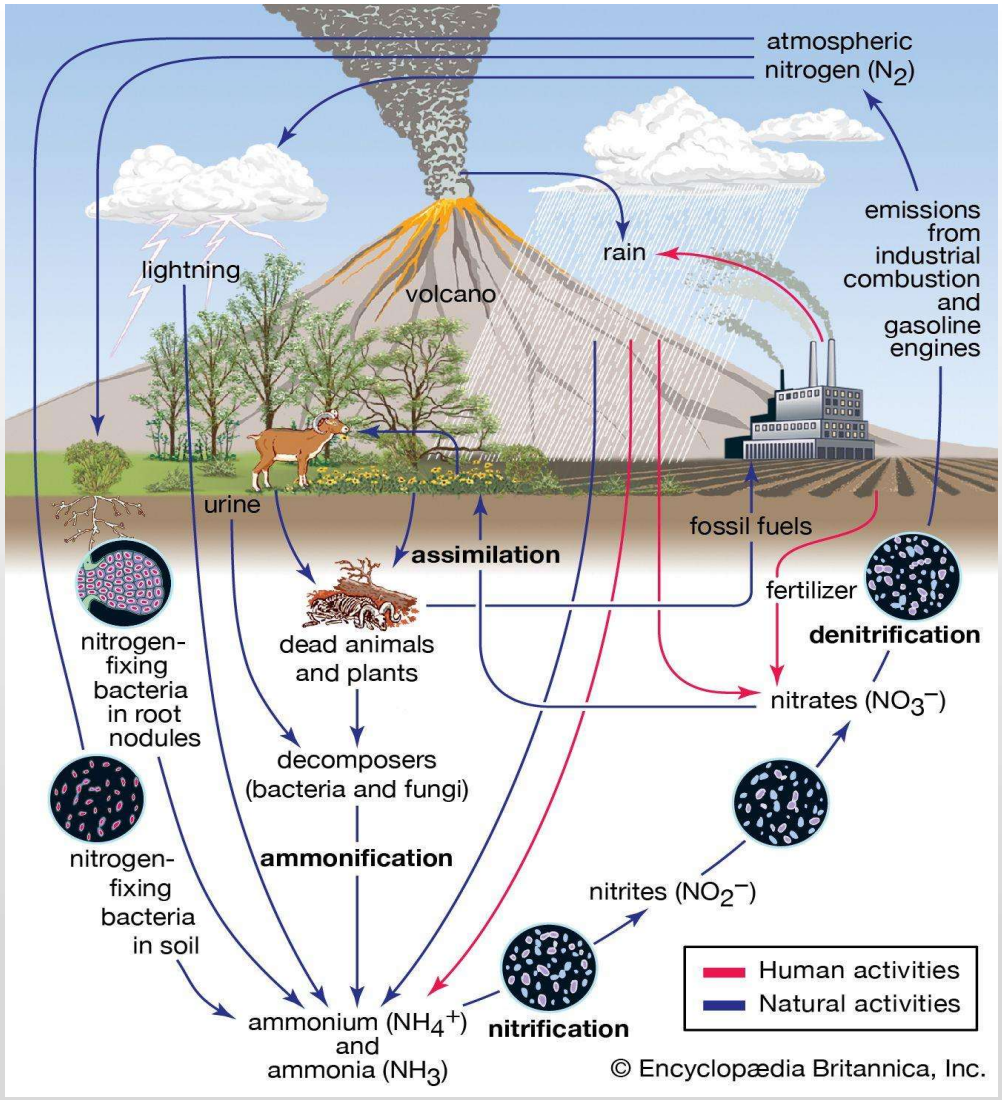
- Carbon dioxide also dissolves directly from the atmosphere into bodies of water (ocean, lakes, etc.), as well as dissolving in precipitation as raindrops fall through the atmosphere.
- When dissolved in water, carbon dioxide reacts with water molecules and forms carbonic acid, which contributes to ocean acidity.
- It can then be absorbed by rocks through weathering. It also can acidify other surfaces it touches or be washed into the ocean.

Nitrogen Cycle



https://www.britannica.com/science/biogeochemical-cycle

Nitrogen Cycle

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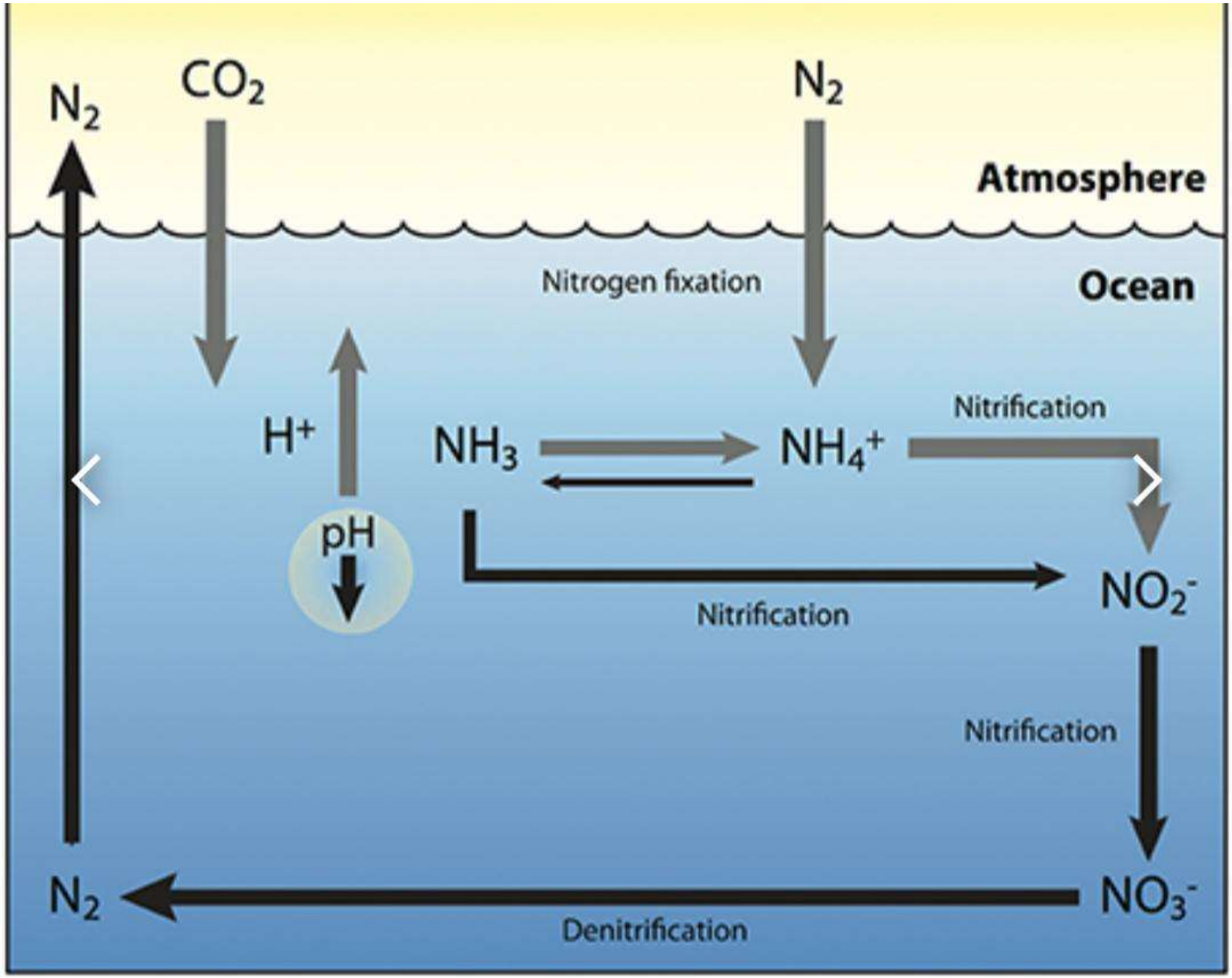
- The **nitrogen cycle** is the biogeochemical cycle by which nitrogen is converted into multiple chemical forms as it circulates among atmospheric, terrestrial, and marine ecosystems.
- Important processes in the nitrogen cycle include fixation, ammonification, nitrification, and denitrification.
- The majority of Earth's atmosphere (78%) is atmospheric nitrogen, making it the largest source of nitrogen.
- However, atmospheric nitrogen has limited availability for biological use, leading to a scarcity of usable nitrogen in many types of ecosystems.

https://www.britannica.com/science/biogeochemical-cycle

Nitrogen Cycle

- The nitrogen cycle is of particular interest to ecologists because nitrogen availability can affect the rate of key ecosystem processes, including primary production and decomposition.
- Human activities such as fossil fuel combustion, use of artificial nitrogen fertilizers, and release of nitrogen in wastewater have dramatically altered the global nitrogen cycle.
- Human modification of the global nitrogen cycle can negatively affect the natural environment system and also human health.


N_2
CO_2
N_2
Atmosphere
Nitrogen fixation
Ocean
H^+
NH_3
NH_4^+
Nitrification
pH
Nitrification
NO_2^-
Nitrification
N_2
Denitrification
NO_3^-

Nitrogen Cycle

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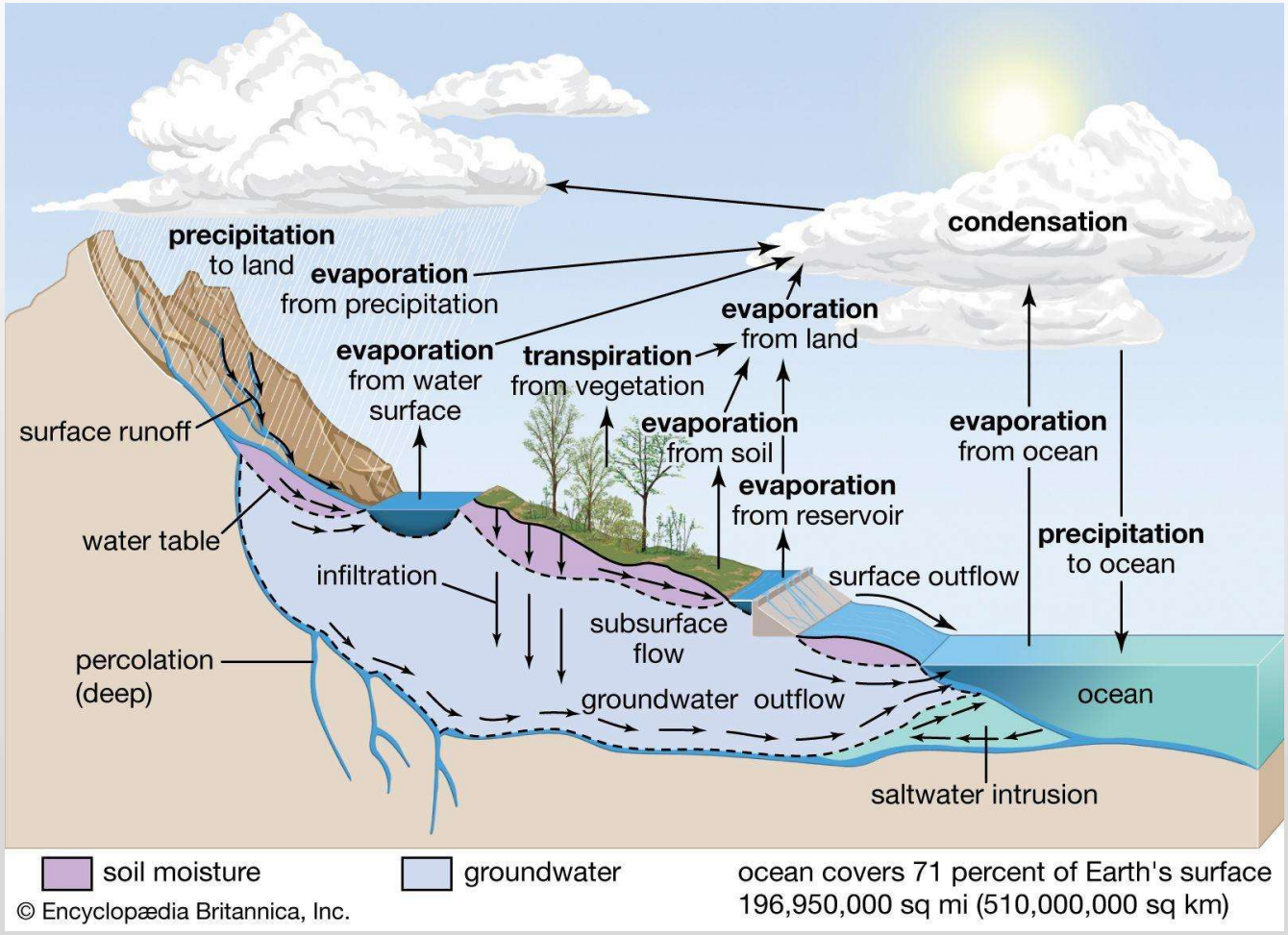
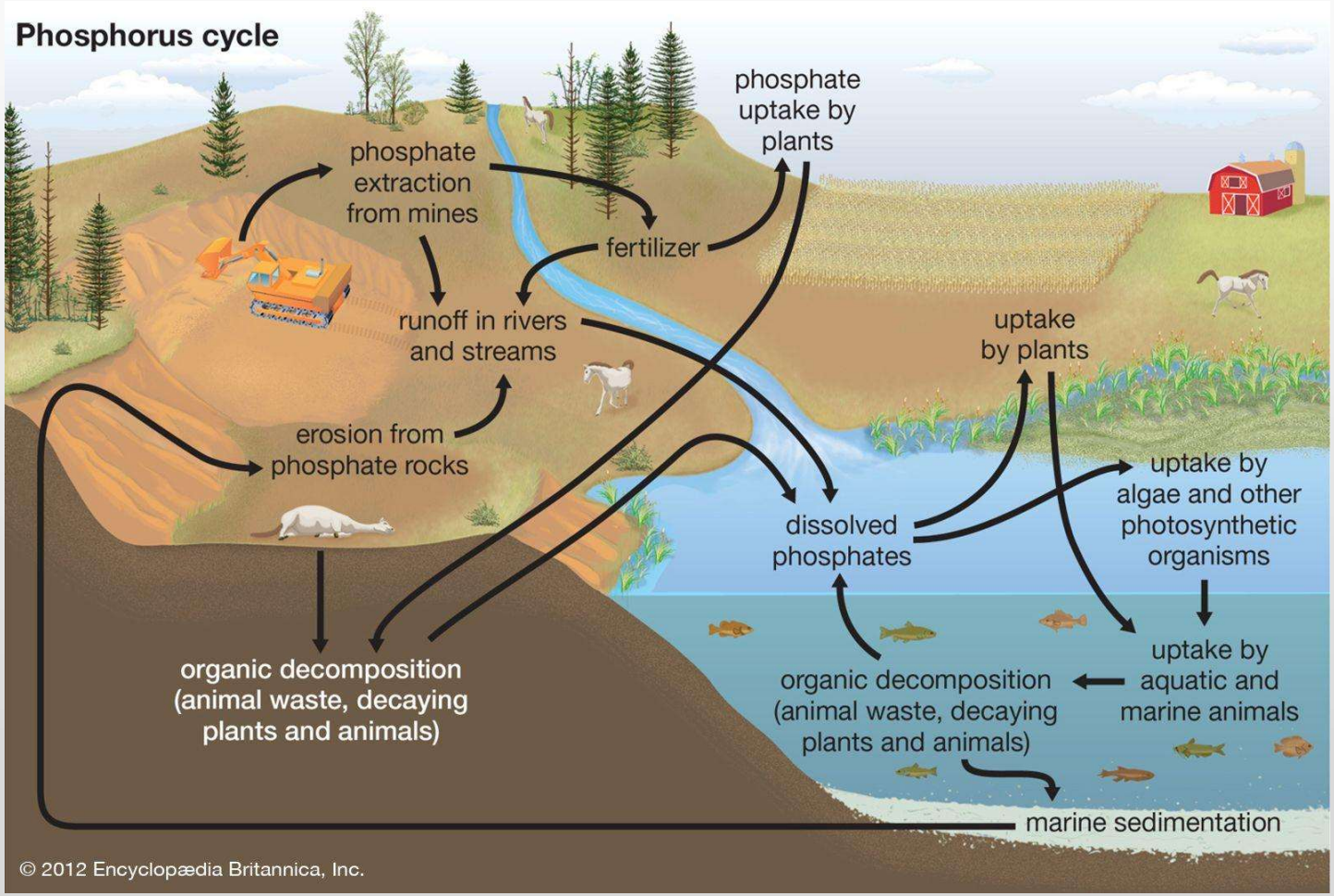
- Nitrogen is present in the environment in a wide variety of chemical forms including organic nitrogen, ammonium (NH^{+4}), nitrite (NO^{-2}), nitrate (NO^{-3}), nitrous oxide (N_2O), nitric oxide (NO) or inorganic nitrogen gas (N_2).
- Organic nitrogen may be in the form of a living organism, humus or in the intermediate products of organic matter decomposition.
- The processes in the nitrogen cycle is to transform nitrogen from one form to another. Many of those processes are carried out by microbes, either in their effort to harvest energy or to accumulate nitrogen in a form needed for their growth.
- For example, the nitrogenous wastes in animal urine are broken down by nitrifying bacteria in the soil to be used by plants.

Hydrologic Cycle



https://www.britannica.com/science/biogeochemical-cycle

Phosphorous Cycle



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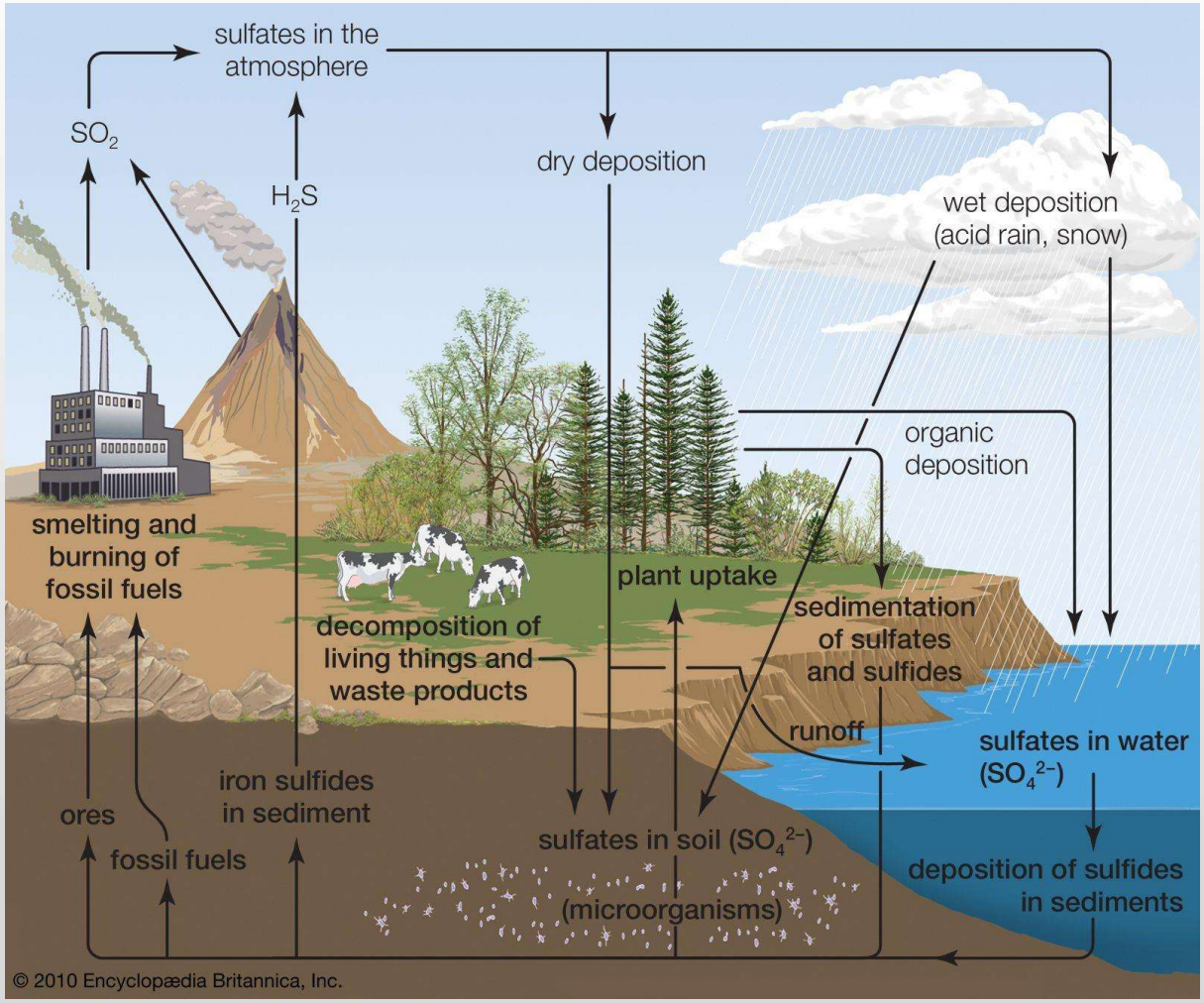
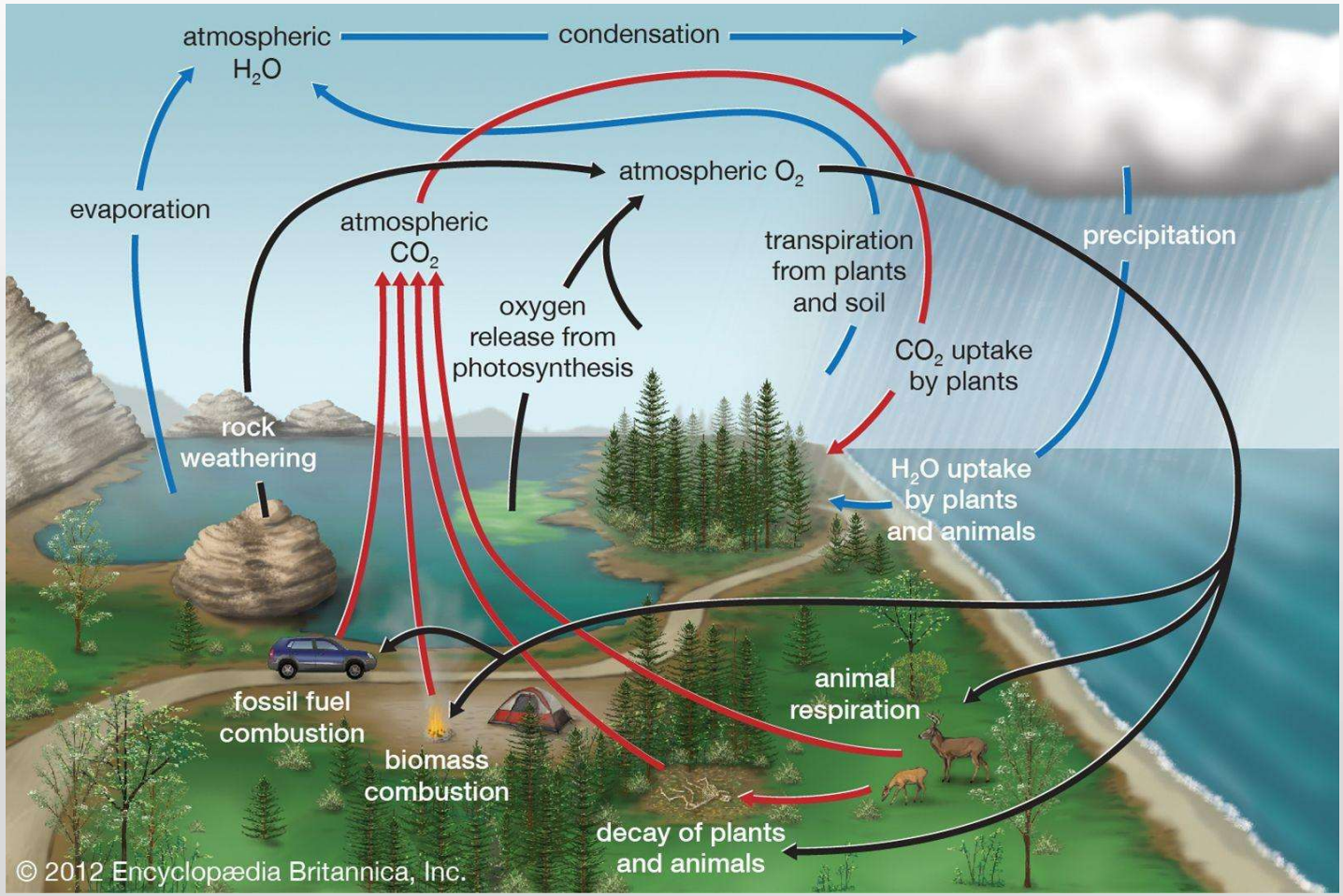
Sulphur Cycle



https://www.britannica.com/science/biogeochemical-cycle

Oxygen Cycle



https://www.britannica.com/science/biogeochemical-cycle

Environmental Gradient

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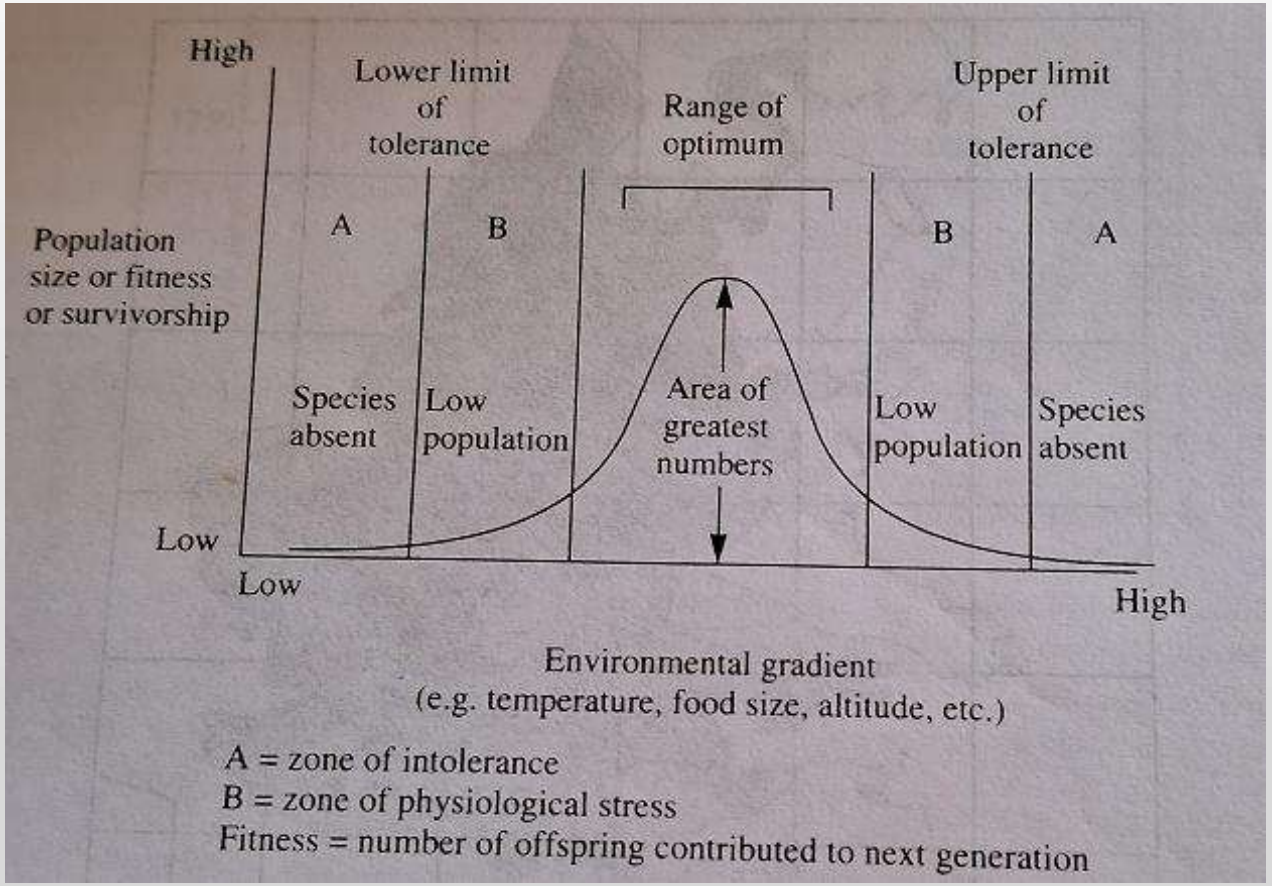
- One of the most obvious pattern in nature is the uneven distribution of organisms over the globe, because each species are not found in every type of habitat or in every part of the word.
- In the environment, various physical and chemical factors like light, temperature and pH (known as conditions), factors the organism actually uses like food, water, shelter, etc., (known as resources) and the presence of other organisms like predators and competitors with which the organisms interact.
- The level of most environmental factors varies over some limited range, with a gradient between the extremes: high to low, large to small. For example, if we examine temperature, there is a global gradient from the equator towards north or south, a regional gradient with altitude, a local gradient within vegetation (from top to bottom).
- To live and thrive in a given situation, each species must have the essential resources and conditions necessary for growth and reproduction.
- Each species only functions efficiently over a limited part of the each environmental gradient-its tolerance range.

A schematic tolerance curve for a single species population existing on a single environmental gradient



➢Areas where no individuals of the species can survive because physical conditions or lack of food resources are too extreme for that species to permit survival- the zone of intolerance.

plot some measure of success of species population (such as population density) against the gradient, tolerance curve is generated, which is also known as "universal tolerance curve".

➢Within the range of optimum of the curve, the majority of individuals of species population can survive and reproduce, thus a large population size is maintained as the conditions along this part of the environmental gradient are ideal for the species.

➢Beyond this range, towards the high and low ends of the environmental gradient, most individuals suffer increasing physiological stress- zone of physiological stress. Here the conditions are less than optimal such that most individuals can stay alive but cannot function efficiently nor reproduce.

Tolerance Levels

- Tolerance refers to the range of physical conditions within which an organism can survive.
- The concept of tolerance allows us to make predictions about where species will be found.
- A tolerance curve, summarizes the responses of organisms to different levels of physical environmental factors such as temperature, moisture, pH, altitude, etc.
- The x-axis (along the bottom) represents the range of the environmental factor.
- The y-axis (along the left side) represents the number of individuals of that species that survive at a given level of the environmental factor.

Tolerance Levels

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- Considering one environmental gradient, if we plot some measure of success of species population (such as population density) against the gradient, tolerance curve is generated, which is also known as “universal tolerance curve”.
- Within the range of optimum of the curve, the majority of individuals of species population can survive and reproduce, thus a large population size is maintained as the conditions along this part of the environmental gradient are ideal for the species.
- Beyond this range, towards the high and low ends of the environmental gradient, most individuals suffer increasing physiological stress- zone of physiological stress. Here the conditions are less than optimal such that most individuals can stay alive but cannot function efficiently nor reproduce.
- Areas where no individuals of the species can survive because physical conditions or lack of food resources are too extreme for that species to permit survival- the zone of intolerance.

Environmental Impact Assessment

EIA is a process that requires consideration of the environment and public participation in the decision making process of project development.
An EIS, Environmental Impact Statement is a review document prepared for assessment in the EIA process.
The stages of EIA include the following:

1. Screening: To decide which project should be subject to environmental assessment. Criteria are threshold, size of project and sensitivity of the environment.
2. Scoping: is the process which defines the key issues that should be included in the environmental assessment.
3. EIS preparation: is the scientific and objective analysis of the scale, significance and importance of the impacts identified.
4. Review: Environmental assessment is reviewed by a government agency or an independent nreview panel. The review panel guides the study and then advises the decesion makers.

Thank You

Name of the Faculty: Dr.Gulmi Chakraborty

Department of Chemistry

Mail Id: gulmi.chakraborty@cgu-odisha.ac.in

www.cgu-odisha.ac.in

C. V. Raman Global University
ODISHA BHUBANESWAR INDIA